

Task 1 — Document 05: System Architecture

Problem Statement ID: PS 26117 / SIH26117 **Title:** Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work **Organization:** Mangalore Refinery and Petrochemicals Limited (MRPL)
Category: Software | **Theme:** Smart Automation

Architectural Overview

The **Sovereign AI Workbench** is structured around a 4-tier modular, air-gapped architecture. Designed for single-node workstation or edge server deployment, the system decouples user interface, backend routing, vector search processing, and local LLM runtime execution.

Visual Diagram Flow: GRAPH TD	
Component / Module	Dark-Themed Single Page App (Vanilla HTML5 / CSS3 / JS)
Component / Module	Knowledge Mode Toggle (ON / OFF)
Component / Module	Multimodal File Upload (PDF, TXT, PNG, JPG, WEBP)
Component / Module	FastAPI Server (main.py) Port: 8000
Component / Module	PDF Parser (pypdf)
Component / Module	Local OCR Engine (GLM-OCR:q8_0)
Component / Module	Semantic Text Chunker (500 char chunks / overlap)
Component / Module	Embeddings Model (nomic-embed-text)
Component / Module	In-Memory Cosine Similarity Search Engine
Component / Module	Local Ollama Service Runtime
Component / Module	qwen3.5:4b (General & Grounded RAG)
Component / Module	qwen2.5-coder:7b (Dedicated Coding)
Component / Module	glm-ocr:q8_0 (Vision OCR Extraction)
Component / Module	data/knowledge_base/ (Source Documents & Images)
Component / Module	data/kb_index.json (Serialized Vector Index)

Architectural Layer Details

1. Presentation Layer (Web UI)

- **Technology:** Single-page application using HTML5, CSS3 (Flexbox/Grid, Dark Modern Palette), and asynchronous JavaScript (**fetch** API).
- **Functionality:**
 - Chat window rendering user prompts and AI responses with markdown support.
 - Interactive file uploader with real-time status notifications.
 - Knowledge ON/OFF toggle controlling RAG augmentation.

- Real-time display of retrieved source documents and cosine similarity scores.

2. Application Gateway Layer (`main.py`)

- **Technology:** Python FastAPI with Uvicorn ASGI server.
- **Functionality:**
 - Serves static UI assets (`frontend/index.html`).
 - Endpoints: **POST** `/upload` for file processing, **POST** `/chat` for conversational Q&A.
- **Task Router:** Inspects prompt text and parameters to select the execution path (Coding Model vs. RAG Grounded Model vs. General LLM).

3. Document Processing & Local Vector RAG Engine

- **Text Parser:** `pypdf` extracts clean text streams from multi-page PDFs.
- **OCR Extractor:** Invokes `glm-ocr:q8_0` via local Ollama API to transcribe scanned images.
- **Chunking Module:** Splits documents into overlapping semantic chunks (500 characters) to preserve contextual boundaries.
- **Embedding Pipeline:** Transmits text chunks to local `nomie-embed-text` model via Ollama to generate 768-dimensional dense vector representations.
- **Vector Storage & Search:** Saves chunk metadata and vectors into `data/kb_index.json`. Executes fast in-memory dot-product cosine similarity search to retrieve top-K matching contexts.

4. Local Model Execution Layer (Ollama)

- **Runtime:** Ollama daemon running on localhost (`http://127.0.0.1:11434`).
- **Model Palette:**
 - `qwen3.5:4b`: Primary general-purpose LLM for conversational chat and grounded RAG synthesis.
 - `qwen2.5-coder:7b`: Code generation specialist.
 - `glm-ocr:q8_0`: Document image OCR specialist.
 - `nomie-embed-text`: High-speed text embedding generator.

5. Storage Layer

- **Physical Disk Storage:** All files reside in local workstation directory `d:\isha\sovereign-ai\data\`.
- **Air-Gap Security:** Zero external network connections; completely self-contained.